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RAVINEESH SINGH



Ravineesh_Singh_Resume.pdf

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Howrah, West Bengal, India - +91 7980561223 - ravineeshsingh23@gmail.com

LinkedIn - GitHub - Portfolio - LeetCode

- Computer Science & **Engineering undergraduate** at IEST Shibpur with an 8.81/10 CGPA, combining strong analytical
- problem solving with technical depth in **Data** Structures & Algorithms, Java and C# development, **Enterprise Data**
- Pipelines, and Applied Machine **Learning**. Experienced in designing scalable **data** ingestion workflows, training deep **learning** architectures (CNNs), and applying structured, secure **engineering** practices to deliver high-impact, **customerfocused** solutions for **enterprise** and financial services environments.

Education *

Indian Institute of Engineering Science and Technology (IEST), **Shibpur**
West Bengal, India

- Bachelor of Technology in Computer Science and **Engineering** - CGPA: 8.81 / 10
Aug 2024 - May 2028
- Higher Secondary **Education** (Class XII) - UPMSF
Uttar Pradesh, India
Uttar Pradesh Board of High **School** and **Intermediate Education** - **Percentage:** 83.6%
Apr 2021 - May 2023
- Secondary **Education** (Class X) - UPMSF
Uttar Pradesh, India
Uttar Pradesh Board of High **School** and **Intermediate Education** - **Percentage:** 89.17%
Apr 2019 - May 2021

Featured Projects *

- Real Estate Price **Predictor** & Market **Analytics** Platform
- **Python**, Pandas, Scikit-learn, SQL **GitHub**
- AI-Driven Precision **Agriculture**: Weed Detection CNN
- **Python**, Keras, **TensorFlow**, Deep Learning **GitHub**

Technical & Professional Skills *

- Python, **Java**, C#, C++, C, SQL, JavaScript (Basics)
- Enterprise Application Development (**Java** / C#), Cloud **Concepts** (AWS / Azure **Fundamentals**), RESTful APIs, ETL Pipelines, Relational Database Design (DBMS)
- **Convolutional** Neural Networks (CNNs), Deep Learning, Predictive Modeling, Regression, **Classification**, Scikit-learn, Keras, TensorFlow
- Pandas, NumPy, Matplotlib, Seaborn, Business Intelligence (Power BI / Tableau **concepts**), Exploratory **Data** Analysis (EDA)
- **Data** Structures & Algorithms (DSA), Object-Oriented Programming (OOP with **Java** / C#), Git, GitHub, VS Code, Linux/Bash, Command Line / Terminal (CLI), Agile / Scrum, SDLC

Certifications & Coursework *

Leadership & Community Engagement *

- Core Member - Icche (Social Welfare Club, IEST Shibpur)
- **IEST Shibpur**
- ETL & **Data** Ingestion Pipeline: Engineered an automated web scraping and ETL pipeline in Python to extract, clean, and normalize 1,000+ multi-attribute property records from 99acres.com, ensuring schema integrity.
- **Feature Engineering** & Machine Learning: Architected end-to-end **data** preprocessing workflows incorporating outlier detection, categorical encoding, **feature** scaling, and multi-algorithm regression (Linear, Ridge, Random Forest) to forecast valuations with high precision.
- Business Intelligence & Market Insights: Developed exploratory **data** analysis (EDA) frameworks to **model** price elasticity, geographical valuation trends, and commercial demand indicators for data-backed decision making.
- Dataset **Engineering** & **Augmentation**: Curated and annotated a domain-specific dataset of 2,000 high-resolution agricultural images, applying image transformation and **augmentation** pipelines (zoom, shear, rotation) to prevent **overfitting**.
- Convolutional Neural Network (CNN): Designed and trained a custom deep CNN architecture in Keras utilizing Batch Normalization and Dropout regularization ($p=0.4$) for high-accuracy **feature** extraction and real-time classification.

- **Model** Optimization & Edge **Deployment**: Benchmarked **model** performance using Precision, Recall, F1-Score, and Confusion Matrix, targeting edge **deployment** to automate targeted herbicide spraying and drive sustainable **agribusiness efficiencies**.

Programming Languages *

- **Cloud & Modern Enterprise**:

Data Science & AI / ML *

Data Analysis & Visualization *

Software Engineering & Tools *

- Command-Line & Terminal Fundamentals: Completed a hands-on Terminal course covering shell navigation, file and directory operations, and core command-line utilities for efficient development workflows (Caddy).
- Data **Structures** & **Algorithms** (DSA): Comprehensive **coursework** in algorithmic problem solving, computational complexity optimization, dynamic **programming**, and tree/graph **structures** (Udemy).
- Core Academic **Coursework**: Data **Structures** & **Algorithms**, Design & Analysis of **Algorithms**, Object-Oriented **Programming**, Database **Management** Systems (DBMS), Operating Systems, Discrete Mathematics.
- **Educational** Outreach & Mentorship: Volunteer 3 days weekly to teach foundational STEM and literacy to underprivileged K-12 students, fostering **educational** equity and measurable academic improvement.
- Project & Event **Management**: Lead cross-functional student volunteer teams to plan and execute campus-wide community welfare drives, logistics for annual Sports Days, and regional clothing/relief campaigns.